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Introduction to Data Communications

Topic 12 –Introduction to Network Security

Our vision is to be Singapore's private education institution of choice, and we are committed to helping our students achieve their career and educational goals through lifelong learning opportunities.

We build futures, one success story at a time.

Integrity . Knowledge . Support . Opportunity . Results



Introduction to Network Security:

1. Network security involves implementing measures to protect the integrity, confidentiality, and availability of data transmitted over a network.
2. It encompasses various technologies, policies, and procedures designed to prevent unauthorized access, data breaches, and other cyber threats.

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Goals of Network Security:

1. **Confidentiality**: Ensuring that data is only accessible to authorized users.
2. **Integrity**: Preventing unauthorized alteration or modification of data.
3. **Availability**: Ensuring that network resources and services are accessible and operational when needed.
4. **Authentication**: Verifying the identity of users and devices accessing the network.
5. **Authorization**: Granting appropriate access privileges to authenticated users.
6. **Non-repudiation**: Ensuring that actions or transactions cannot be denied by the parties involved.

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TCP/IP Model-based Vulnerabilities

The TCP/IP model, which includes the layers of the Internet Protocol Suite, is susceptible to various vulnerabilities that can be exploited by attackers to compromise network security.

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IP Vulnerabilities

1. IP (Internet Protocol) vulnerabilities include weaknesses in the IP packet structure and routing mechanisms.
2. Man-in-the-Middle (MITM) Attack: An attacker intercepts and possibly alters communication between two parties without their knowledge.
3. Denial of Service (DoS) Attack: An attack that aims to disrupt or disable network services or resources, making them unavailable to legitimate users.

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DoS Attack / Types:

1. DoS attacks overload network resources or services to the point of exhaustion, causing disruption or denial of service.
2. Types include:
    1. **UDP Flooding**: Floods the target with a large volume of User Datagram Protocol (UDP) packets.
    2. **TCP Syn Flood**: Exploits the TCP handshake process by sending a flood of SYN packets without completing the handshake.
    3. **TCP Session Hijacking**: Hijacks an established TCP session between two parties by intercepting and injecting malicious packets.
    4. **TCP Reset Attack**: Spoofs TCP reset packets to terminate legitimate connections.

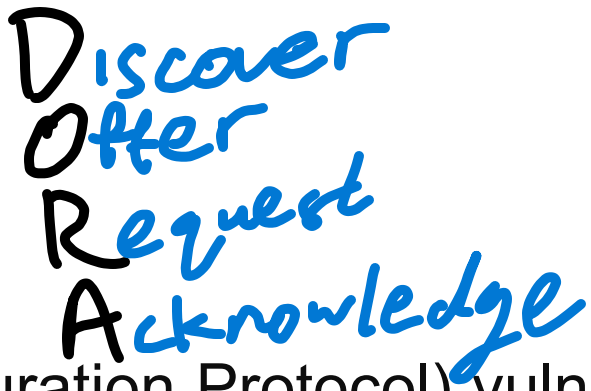
ICMP Vulnerabilities:



1. ICMP (Internet Control Message Protocol) vulnerabilities involve exploiting weaknesses in ICMP messages and functionalities.
2. ICMP Sweep Attack: Scans a range of IP addresses to identify active hosts. (FIREWALL)
3. Traceroute Attack: Determines the route and network topology of a target by exploiting ICMP error messages.
4. ICMP Tunneling: Conceals unauthorized network traffic within ICMP packets.
5. ICMP Smurf Attack: Amplifies a DoS attack by broadcasting ICMP echo requests to a network's broadcast address.

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DHCP Vulnerabilities



- DHCP (Dynamic Host Configuration Protocol) vulnerabilities include attacks that exploit weaknesses in DHCP message exchanges.
- DHCP Starvation: Exhausts available IP addresses in a DHCP pool by requesting multiple leases.
- DHCP Spoofing: Impersonates a legitimate DHCP server to provide false configuration information to clients

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App Layer Vulnerabilities

1. Application layer vulnerabilities refer to weaknesses in software applications that can be exploited by attackers to compromise security.
2. These vulnerabilities often arise from flaws in application design, implementation, or configuration.

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DNS Spoofing



1. DNS Spoofing, also known as DNS cache poisoning, involves corrupting the DNS cache of a DNS server to redirect DNS queries to malicious websites.
2. Attackers exploit vulnerabilities in DNS protocols or servers to inject false DNS records into the cache, leading to DNS resolution errors and redirection of traffic to malicious sites.

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Email Spoofing

- Email Spoofing involves forging the header information of an email message to make it appear as if it originated from a different sender.

- Attackers use email spoofing to impersonate trusted entities, such as banks or government agencies, to trick recipients into disclosing sensitive information or clicking on malicious links

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Spoofing Attacks Covered In Summary

1. Spoofing attacks involve impersonating legitimate entities or devices to deceive users or systems.
2. Types of spoofing attacks include:

LAYER 2
   1. ARP Spoofing: Manipulating ARP (Address Resolution Protocol) tables to associate a different MAC address with an IP address.

LAYER 3
   2. IP Spoofing: Falsifying the source IP address in IP packets to impersonate another host.
   3. DHCP Spoofing: Impersonating a DHCP server to distribute false IP configuration information to clients.

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Intranet Network Security

- Intranet security focuses on protecting internal network resources, systems, and data from unauthorized access, misuse, or attacks.
- Key elements of intranet security include:
  - **Intrusion Detection System (IDS)**: Monitors network traffic for suspicious activity or known attack patterns and alerts administrators of potential security incidents.
  - **Intrusion Prevention System (IPS)**: Proactively blocks or mitigates detected threats to prevent them from reaching their targets.
  - **Honeynet and DMZ (Demilitarized Zone)**: Honeypots and honeynets are decoy systems or networks designed to lure and deceive attackers, while a DMZ is a network segment that separates internal and external networks, hosting public-facing services.

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Intranet Network Security

- **Proxy Server**: Acts as an intermediary between clients and servers, filtering and caching web content to improve performance and security.
- **Firewall**: Inspects and controls incoming and outgoing network traffic based on predetermined security rules to enforce access policies and protect against unauthorized access.
- **Router ACL (Access Control List)**: Filters network traffic based on IP addresses, port numbers, or other criteria specified in access control lists configured on routers.
- **VPN (Virtual Private Network)**: Establishes secure, encrypted connections over public networks to enable remote users to access internal network resources securely

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AAA Security Framework

1. The AAA framework stands for Authentication, Authorization, and Accounting, which are three core components of network security.
2. AAA ensures that only authorized users can access network resources, and their activities are tracked for auditing and accountability purposes.

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Intro. to AAA Framework

1. **Authentication**: Verifies the identity of users or devices attempting to access network resources, typically through credentials such as usernames and passwords, biometrics, or digital certificates.
2. **Authorization**: Determines what actions or resources users are allowed to access based on their authenticated identity and assigned permissions.
3. **Accounting**: Tracks and records user activities and resource usage for auditing, billing, or regulatory compliance purposes.

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AAA implementation - Radius

1. RADIUS (Remote Authentication Dial-In User Service) is a networking protocol that provides centralized authentication, authorization, and accounting for remote access users.
2. It is commonly used in dial-up and VPN connections, as well as wireless networks, to authenticate users and authorize their access to network resources.

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AAA implementation - TACACS+

1. TACACS+ (Terminal Access Controller Access-Control System Plus) is a security protocol developed by Cisco for controlling access to network devices.
2. It separates authentication, authorization, and accounting processes into separate stages, providing more granular control over access policies and user privileges.

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Comparison (Radius vs TACACS+)

- **RADIUS**:
  - Widely supported across various network devices and platforms.
  - Primarily used for authentication and accounting, with limited authorization capabilities.
  - Relies on UDP for communication and is vulnerable to packet interception and replay attacks.
- **TACACS+**:
  - Developed by Cisco and offers more comprehensive authentication, authorization, and accounting capabilities.
  - Provides separate authentication and authorization processes, allowing for more fine-grained access control.
  - Uses TCP for communication, offering better reliability and security compared to RADIUS

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Encryption:

1. Encryption is the process of encoding data to make it unreadable to unauthorized parties, ensuring confidentiality and data integrity.
2. Two main types of encryption include symmetric key encryption and asymmetric key encryption.

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Symmetric Key Encryption

- Uses a single key for both encryption and decryption.

- Examples include DES (Data Encryption Standard), AES (Advanced Encryption Standard), and 3DES (Triple Data Encryption Standard).

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Asymmetric Key Encryption

1. Uses a pair of keys: a public key for encryption and a private key for decryption.
2. Examples include RSA (Rivest-Shamir-Adleman) and ECC (Elliptic Curve Cryptography).

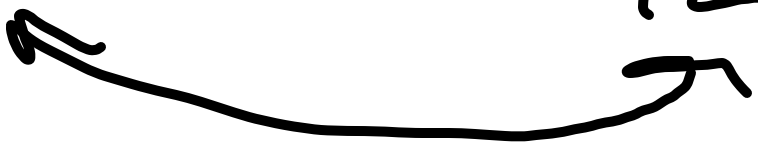
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Digital Signature

1. A digital signature is a cryptographic technique used to verify the authenticity and integrity of digital messages or documents.
2. It involves creating a unique digital fingerprint of the data using a private key and attaching it to the message.

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Digital Certificate / SSL Certificate: TLS



- A digital certificate is an electronic document that verifies the identity of an entity, such as a website or individual.
- SSL (Secure Sockets Layer) certificates are a type of digital certificate used to secure communication between web browsers and servers, providing encryption and authentication

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General Concepts

1. **Security Management**: Involves establishing policies, standards, guidelines, and procedures to manage and mitigate security risks effectively.
2. **Disaster Recovery**: Plans and processes for restoring operations and data in the event of a catastrophic failure or security breach.
3. **Attack Mitigation Plan**: Strategies and measures to detect, prevent, and respond to security threats and attacks.
4. **Whois Database**: A publicly accessible database containing information about registered domain names and their owners, used for network administration and security purposes.

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Phishing Attacks

Overview: Phishing involves deceiving individuals into disclosing confidential information, such as passwords or financial details, often via fake emails or websites that appear legitimate.

Prevention:

Staff Education: Regularly train employees to spot and avoid phishing scams.

Email Security Filters: Deploy sophisticated email filters to block potentially harmful messages.

Multi-Factor Authentication (MFA): Implement extra security steps, like SMS codes or biometric verification, to reduce the misuse of compromised accounts.

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Distributed Denial of Service (DDoS) Attacks

Overview: A DDoS attack overwhelms a server with massive traffic, rendering it unavailable to legitimate users.

Prevention:

Traffic Filtering: Use DDoS protection services that block malicious traffic before it impacts servers.

Load Balancing: Spread network traffic across multiple servers to prevent any one server from being overloaded.

Rate Limiting: Restrict the number of requests a server can process over a set period.

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SQL Injection

Overview: SQL injection exploits weaknesses in web applications by inserting malicious SQL queries to access or manipulate the database.

Prevention:

Sanitize Input: Ensure all user inputs are validated and sanitized before processing.

Parameterized Queries: Use prepared statements and parameterized queries to block unauthorized code.

Security Audits: Conduct regular security audits to detect and resolve vulnerabilities in web applications.

This version maintains the key details while presenting the information in a slightly different way.

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Malware (Malicious Software)

Overview: Malware is designed to disrupt or gain unauthorized access to systems, including threats like viruses, worms, and ransomware.

Prevention:

Use Antivirus Software: Maintain up-to-date antivirus and anti-malware tools.

Regular Patching: Ensure software and systems are updated regularly to fix security weaknesses.

Data Backups: Routinely back up important data to mitigate damage from malware, especially ransomware.

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Man-in-the-Middle (MitM) Attacks

Overview: In MitM attacks, a hacker intercepts and alters communication between two parties who think they are communicating securely.

Prevention:

Encryption: Implement end-to-end encryption (SSL/TLS) to secure data transmissions.

Use Secure Wi-Fi: Avoid conducting sensitive activities on public Wi-Fi, or use a VPN for secure access.

Strong Authentication: Use robust authentication methods to verify the legitimacy of the parties involved in the communication.

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Thank you

